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Prn = 202501080014

Dataset = Movie review

20. Problem Statement

NUMPY

1. Addition

```
[2]
result_add = phrase + sentiment
print(result_add[:10])

[156064 156065 156065 156067 156068 156069 156070 156070 156072 156072]
```

2.Subtraction

```
[3]
result_sub = phrase - sentiment
print(result_sub[:10])

[156058 156059 156061 156061 156062 156063 156064 156066 156066 156068]
```

3.Multiplication

```
result_mul = phrase * sentiment
print(result_mul[:10])

[468183 468186 312126 468192 468195 468198 468201 312136 468207 312140]
```

4.Division

```
[5]
result_div = phrase / (sentiment + 1)  # avoid divide by zero
print(result_div[:10])

[39015.25       39015.5        52021.         39016.
 39016.25       39016.5        39016.75       52022.66666667
 39017.25       52023.33333333]
```

5. Mean(average)

```
[6]
print("Mean sentiment:", np.mean(sentiment))

Mean sentiment: 2.058875882459422
```

6.Maximum value

```
print("Max sentiment:", np.max(sentiment))
```

```
Max sentiment: 4
```

7.Minimum value

```
print("Min sentiment:", np.min(sentiment))
```

```
Min sentiment: 0
```

8.standard deviation

```
[9]
print("Std Dev:", np.std(sentiment))
```

```
Std Dev: 0.7392832098561429
```

9.Square values

```
[10]
square = np.square(sentiment)
print(square[:10])
```

```
[9 9 4 9 9 9 9 4 9 4]
```

10.sort values

```
[11]
sorted_values = np.sort(sentiment)
print(sorted_values[:10])
```

```
[0 0 0 0 0 0 0 0 0 0]
```

Pandas

11.Load dataset and show data

```
[16]
import pandas as pd
df = pd.read_csv("blend.csv")
df
```

	PhraseId	Sentiment
0	156061	3
1	156062	3
2	156063	2
3	156064	3
4	156065	3
...	...	...
66287	222348	1
66288	222349	1
66289	222350	1
66290	222351	1
66291	222352	1

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12. First 10 rows

```
[17]
print(df.head(10))
```

```
   PhraseId  Sentiment
0    156061          3
1    156062          3
2    156063          2
3    156064          3
4    156065          3
5    156066          3
6    156067          3
7    156068          2
8    156069          3
9    156070          2
```

13. Total number of reviews

```
[18]
print("Total reviews:", len(df))

Total reviews: 66292
```

14. Column names

```
[20]
print(df.columns)

Index(['PhraseId', 'Sentiment'], dtype='object')
```

15. Data types

```
[21]
print(df.dtypes)

PhraseId     int64
Sentiment    int64
dtype: object
```

16. Missing values

```
[22]
print(df.isnull().sum())

PhraseId     0
Sentiment    0
dtype: int64
```

17. Count sentiment

```
print(df['Sentiment'].value_counts())

Sentiment
2    38237
3    13550
1    11703
4     1915
0      887
Name: count, dtype: int64
```

18.Average rating

```
df.columns = df.columns.str.strip().str.lower()

print(df.columns)
```

```
Index(['phraseid', 'sentiment'], dtype='object')
```

19. Unique sentiments

```
print(df['sentiment'].unique())
```

```
[3 2 1 4 0]
```

20. Percentage of each sentiment

```
print(df['sentiment'].value_counts(normalize=True) * 100)
```

```
sentiment
2    57.679660
3    20.439872
1    17.653714
4     2.888735
0     1.338020
Name: proportion, dtype: float64
```